Paulson 2020 Merkel cell carcinoma — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Harmony (sample, θ=10) collapses the 3 tumour samples onto ONE continuous component (100%, 0 detached); per-neighbourhood iLISI 1.0→2.25 (cross-patient 1.75/2); every sample traverses the shared embryonic axis end-to-end (|ρ(axis,Crest)|≈0.87, span 100%) — mixed & continuous, not a per-patient archipelago.

Gate 2 — module geography → GO (non-branching)

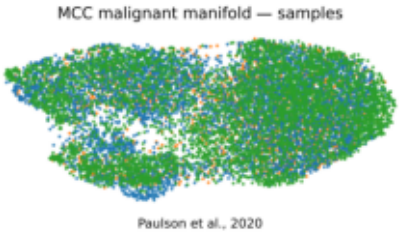
Gast & pSom co-vary along one axis (smoothed r=+0.38; joint-high enrichment 1.22 — no two-arm divergence); Crest is an inverse TERMINAL gradient (pSom↔Crest r=−0.70; Crest-high decile mean pt=0.96) peaking at the terminus, not a Gast arm — no inversion flag.

Gate 3 — cross-repression → NA

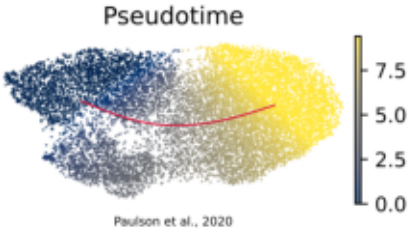
Branching-only. No diverging Gast/pSom hub (single fold-free curve, 0 forks) — no bifurcation to test, so Gate 3 does not apply.

STEP FIGURES

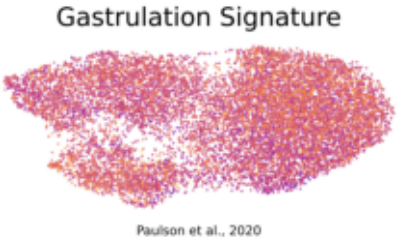
Manifold — samples



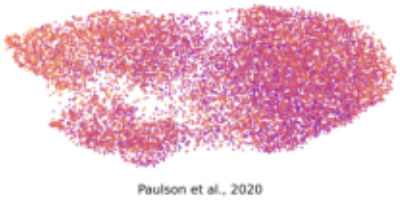
Pseudotime



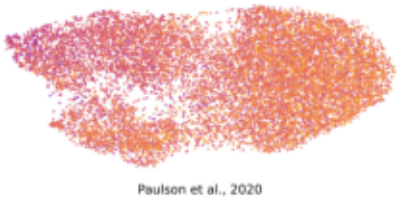
Gastrulation



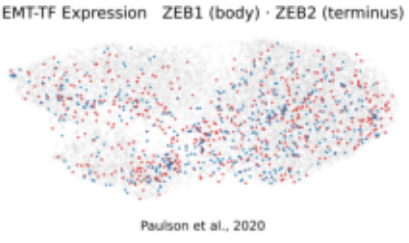
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Merkel cell carcinoma (neuroendocrine skin) (Paulson et al., 2020), 10x — 12,124 malignant cells (3 tumours, 2 patients); non-branching Linear-pSom/Crest.

Verdict.
Pattern reproduces (non-branching): a single continuous gastrulation/somitogenesis → terminal neural-crest axis, rooted at the Gast/pSom-high origin. Gate 1 GO / Gate 2 GO / Gate 3 NA. EMT-TFs ZEB1 (body) · ZEB2 (terminus), both origin/body-loaded; only factors >3% floor.

Parameter choices

Min-cell threshold	≥100 floor (smallest sample 2,031); 3/3 kept, 12,124 cells	**Integration**	Harmony on sample θ=10 (swept 2/5/10); one continuous component
n_pcs / spread	20 / 1.3 (PC1=0.23 CNV-dominated)	**Overdispersed genes**	2,075 (scFates find_overdispersed)
Trajectory	scFates curve N=20, λ=0.3, μ=100 (2 tips, 0 forks, SA 2.83)	**Root**	tip 2 = Gast/pSom-high, Crest-low (Crest is late)
Module means (raw)	Gast +0.037 · pSom +0.051 · Crest −0.006	**EMT-TFs shown**	ZEB1 (14.4%) · ZEB2 (19.1%); both origin/body-loaded